CROP CLASSIFICATION USING SATELLITE IMAGES FULL REPORT

1. Introduction

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Agriculture has always been the cornerstone of human civilization, providing food, raw materials, and livelihoods for billions of people worldwide. In developing countries such as India, agriculture not only ensures food security but also plays a significant role in economic development. With increasing population pressures, climate variability, and limited resources, there is a growing demand for accurate, timely, and large-scale monitoring of agricultural activities. One of the most crucial aspects of agricultural monitoring is **crop classification**, which involves identifying, mapping, and categorizing different crop types across a specific geographic region. Crop classification serves multiple purposes in the agricultural domain. It enables policymakers and researchers to estimate crop acreage, analyze cropping patterns, assess seasonal variability, and forecast yields. For farmers, it provides valuable insights into crop health, stress conditions, and management practices. Governments and agencies also rely on accurate crop classification to plan procurement, trade policies, and subsidies, ensuring effective distribution of resources and food supply management. Traditionally, crop classification was carried out through field surveys and manual data collection. While effective at a local scale, these methods are time-consuming, labor-intensive, and often prone to errors. With the advent of modern technologies, especially **remote sensing and satellite imagery**, the process has undergone a remarkable transformation. High-resolution multispectral and hyperspectral satellite data allow large-scale, repeatable, and objective monitoring of crops over vast geographical areas. When combined with **geographic information systems (GIS), machine learning, and deep learning algorithms**, these technologies enhance the accuracy and reliability of crop classification.In recent years, the use of **vegetation indices** such as the Normalized Difference Vegetation Index (NDVI), texture features, and temporal growth patterns derived from satellite images has further

strengthened crop identification processes. These advancements not only improve precision agriculture but also help address global challenges like climate change adaptation, water resource management, and food security planning.Thus, crop classification has evolved from a manual, small-scale practice to an advanced, technology-driven process that supports sustainable agricultural management. Its significance extends beyond local farming communities to national and global levels, making it a vital component of modern agricultural research and policy formulation.

1.2 Problem Description

Despite the critical role of agriculture in ensuring food security and economic stability, monitoring and classifying crops at large scales remain a challenge. Traditional field-based surveys are labor-intensive, time-consuming, and limited in spatial coverage, leading to delays and inaccuracies in crop data. While satellite imagery provides a promising alternative, variability in spectral signatures, seasonal changes, and atmospheric disturbances introduce complexities in accurate classification. Furthermore, the lack of a fully automated, scalable, and cost-effective framework limits the transformation of raw satellite data into actionable insights, thereby hampering timely agricultural planning, yield forecasting, and resource allocation. Hence, there is a pressing need for a robust and technology-driven crop classification system that leverages satellite imagery with advanced computational techniques to deliver accurate, timely, and scalable solutions for sustainable agricultural management.

1.3 Problem Statement

Design and develop a machine learning-based system for crop classification using satellite images to achieve accurate, automated, and large-scale crop monitoring. Accurate crop classification is essential for effective agricultural planning, yield

forecasting, and resource management. However, traditional field surveys are slow, costly, and limited in coverage, while satellite imagery, though promising, faces challenges such as spectral variability, seasonal changes, and cloud interference. The absence of an automated, scalable, and cost-efficient framework restricts the timely conversion of satellite data into actionable insights. Therefore, there is a need for a robust crop classification system that integrates satellite imagery with advanced machine learning and deep learning techniques to enable precise, timely, and large-scale agricultural monitoring.

1.4 Objectives

• Develop an automated framework for accurate classification of different crop types using satellite imagery.

• Preprocess and enhance satellite data by removing noise, clouds, and distortions for improved feature extraction.

• Apply machine learning and deep learning techniques for effective crop identification and classification.

• Evaluate classification accuracy using ground truth data and satellite imagery.

• Provide a scalable and cost-efficient solution supporting agricultural planning and sustainable management.

1.5 Scope

The proposed system aims to classify crops automatically using satellite images and machine learning models. It will process multi-spectral images, extract relevant features, and classify crops across regions. The application supports farmers, researchers, and policymakers in resource management, yield forecasting, and planning.

2. Overall Description

2.1 Product Perspective

The product will be a software system integrating remote sensing data analysis with machine learning. It consists of modules for data collection, preprocessing, classification, and evaluation. It will utilize tools like Google Earth Engine, QGIS, and Python libraries such as TensorFlow and Scikit-learn.

2.1.1 Hardware Requirements

• Processor: Intel i5 or above
• RAM: 8 GB minimum
• Hard Disk: 500 GB minimum
• GPU: Optional for deep learning models

2.1.2 Software Requirements

• Operating System: Windows / Linux
• Google Earth Engine (GEE)
• QGIS with Semi-Automatic Classification Plugin
• Python with OpenCV, TensorFlow/Keras, and Scikit-learn

2.2 Product Functions

• **Collect and manage** satellite imagery datasets from sources like **Sentinel** and **Landsat** for specific agricultural regions and time periods.

• **Preprocess images** to remove noise, clouds, and atmospheric distortions, ensuring high-quality data for analysis.

• **Perform time-series analysis** on multi-temporal satellite images to monitor crop growth patterns, detect phenological changes, and improve classification accuracy across seasons.

• **Apply machine learning models** such as **Support Vector Machine (SVM)**, **Random Forest (RF)**, and **Convolutional Neural Networks (CNN)** for accurate crop classification.

• **Evaluate classification performance** using accuracy metrics such as **Precision, Recall, F1-Score, and Overall Accuracy** to ensure model reliability.

• **Display and visualize classified outputs** through color-coded maps, charts, and dashboards for easy interpretation and decision-making by users and policymakers.

3. Specific Requirements

3.1 External Interfaces

Inputs: Satellite imagery data, ground truth datasets.
Outputs: Classified crop maps, accuracy reports, and performance metrics.

3.2 Functional Requirements

• The system shall collect satellite images from designated sources.

• The system shall preprocess images to enhance clarity and reduce distortion.

• The system shall classify crops based on spectral and temporal patterns using ML/DL models.

• The system shall evaluate classification performance using statistical metrics.

• The system shall generate visual maps and reports for user interpretation.

3.3 Performance Requirements

The system should process images efficiently with minimal latency, supporting batch operations. It should handle large datasets, maintain high accuracy (>90%), and scale across multiple regions.

3.4 Design Constraints

Limited training data, cloud cover interference, and computational resource constraints may affect performance. System design must ensure compatibility with open-source data and software tools.

3.5 Logical Database Requirements

The proposed system utilizes PostgreSQL as the primary database management system, integrated with the PostGIS spatial extension to efficiently handle geospatial data. The database is designed to store and manage various components of the crop classification workflow, including metadata of satellite images, extracted features, classification outputs, and evaluation metrics.

3.6 Software System Attributes

3.6.1 Reliability

The system shall ensure reliable and consistent crop classification results under varying conditions and datasets.
It should maintain data integrity and deliver repeatable outcomes across multiple runs.

Regular validation and error-handling mechanisms will minimize classification inaccuracies and system failures.

3.6.2 Availability

The system shall be designed for continuous availability, ensuring at least 99% uptime for data analysis and reporting operations.
Scheduled maintenance and updates will be managed with minimal downtime.
It should be capable of recovering quickly from unexpected interruptions or data source failures.

3.6.3 Security

System access shall be restricted to authenticated and authorized users through secure login credentials.
Data transmission between modules must be encrypted to prevent unauthorized access or tampering.
Regular audits and access logs will help monitor and protect sensitive datasets and model results.

3.6.4 Maintainability

The software architecture shall support modular design, allowing for easy updates, bug fixes, and enhancements.
Machine learning models can be retrained or replaced without affecting other components.
Proper documentation and version control will ensure that maintenance and scalability remain efficient.

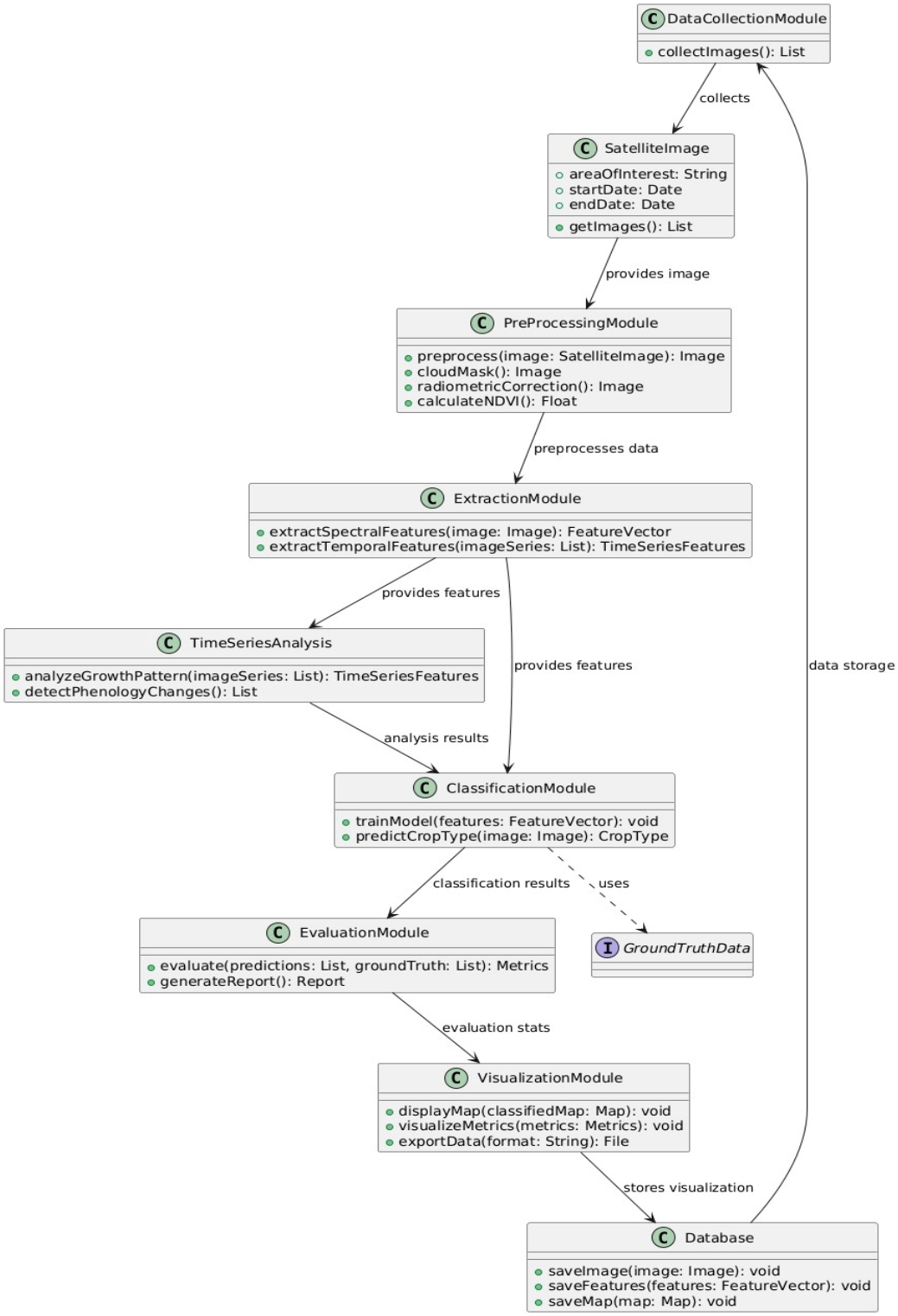
3.6.5 Portability

The system shall be platform-independent and easily deployable across Windows, Linux, and cloud environments.
All dependencies and libraries will be open-source or cross-compatible to ensure smooth portability.
Containerization tools (e.g., Docker) or cloud integration may be used to simplify deployment and scalability.

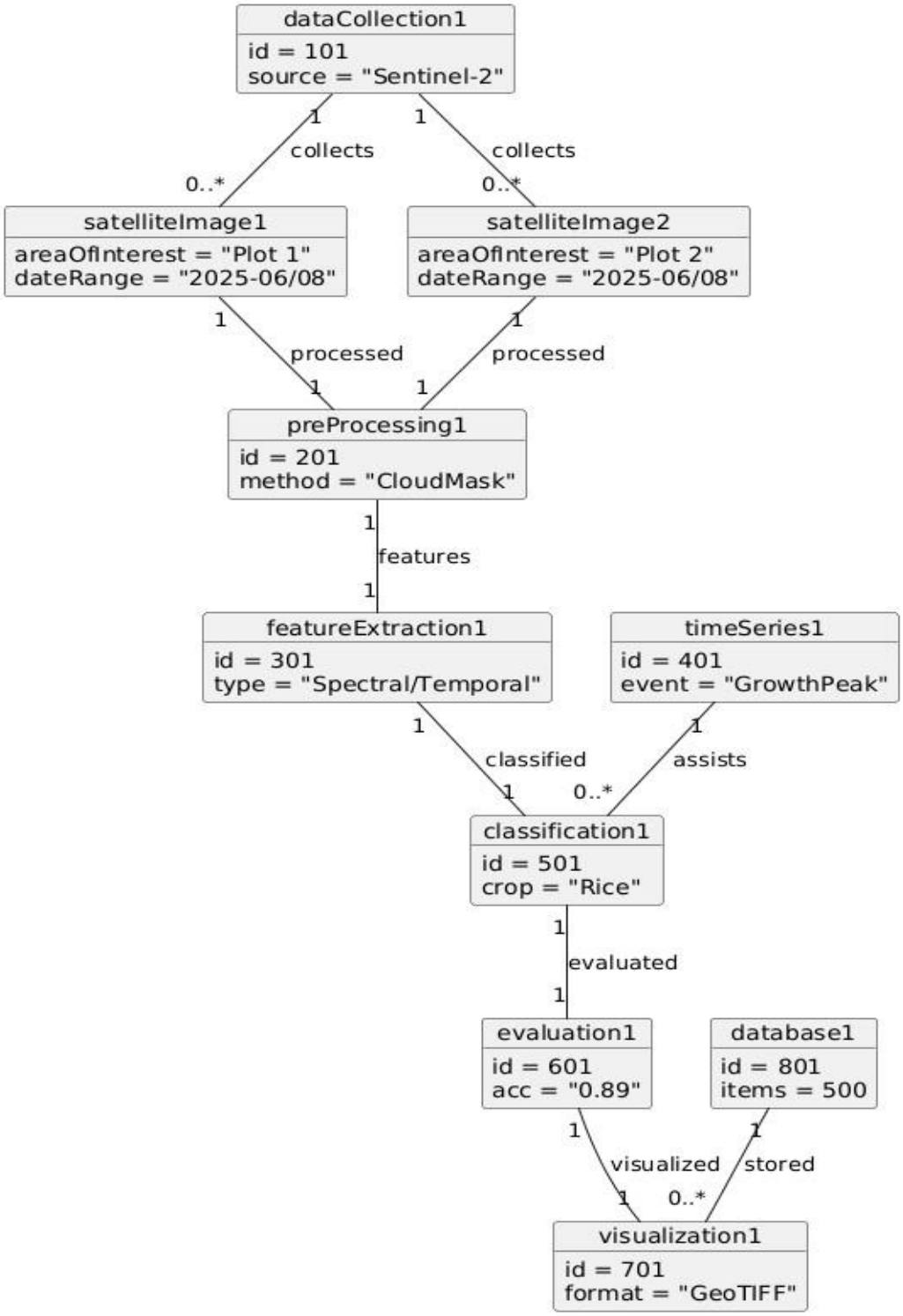
4. Software Design Document

4.1. Structural Design

a. Class Diagrams



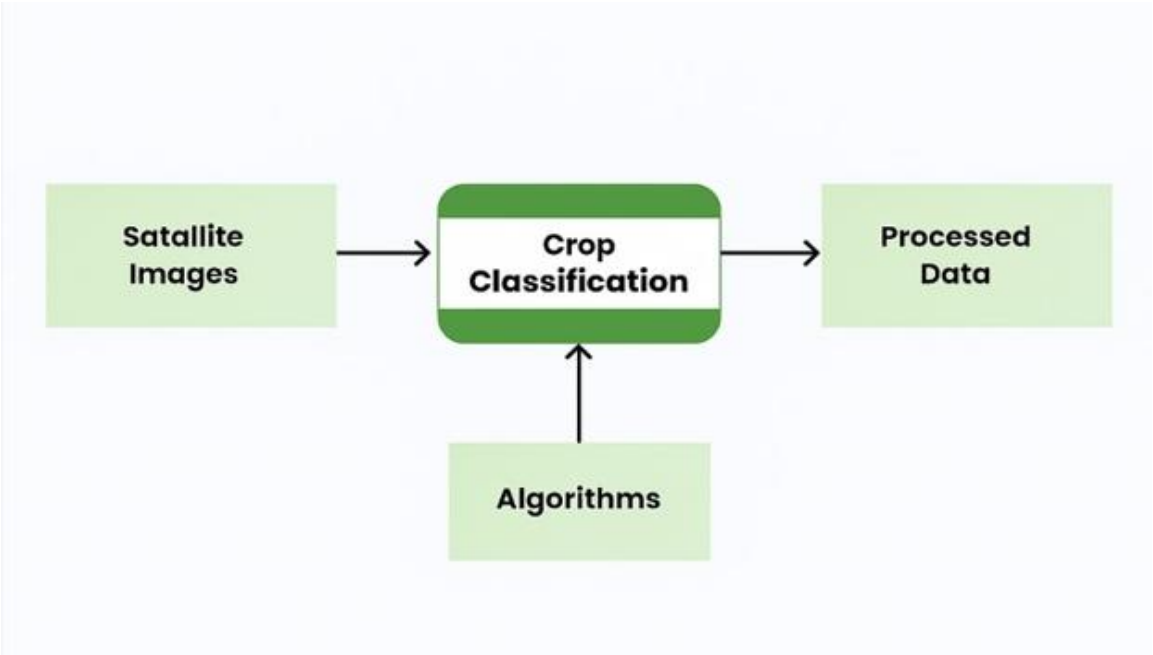
b. Object Diagrams



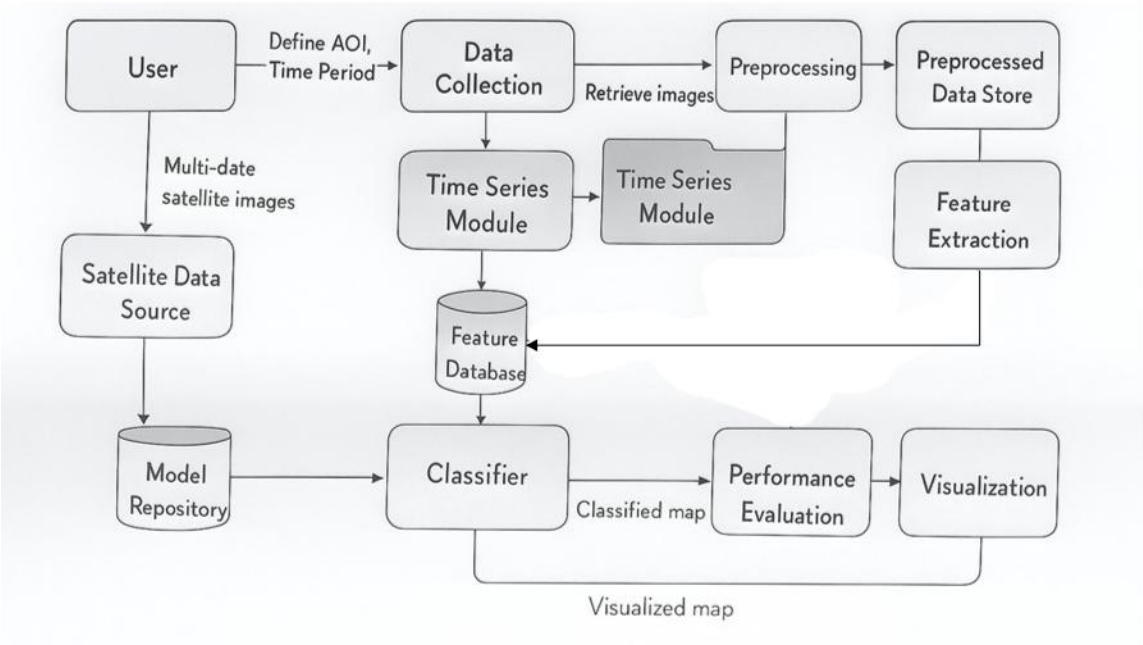
4.2. Data Design

a. Data-flow Diagram

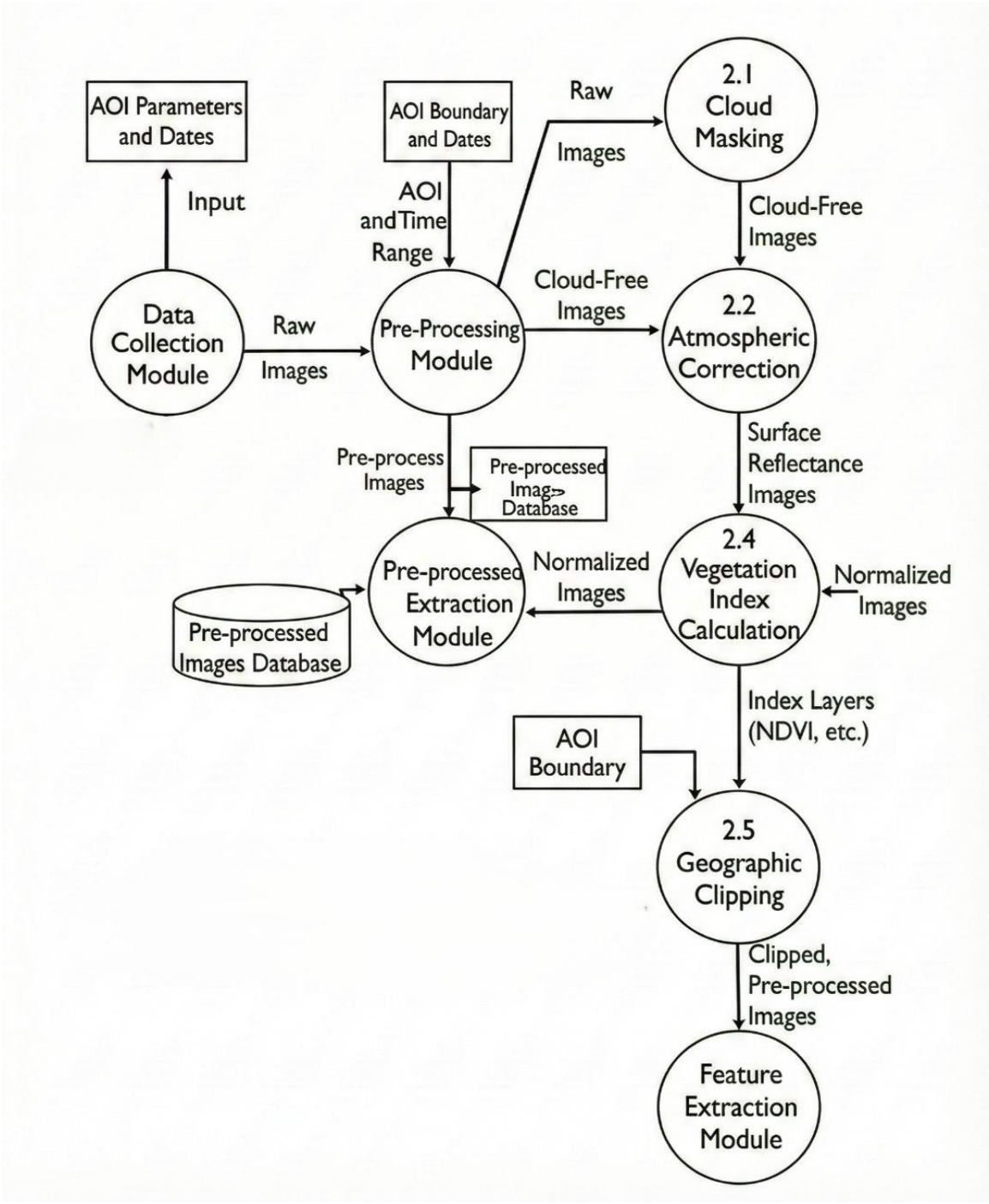
Level 0 DFD:



Level 1 DFD:

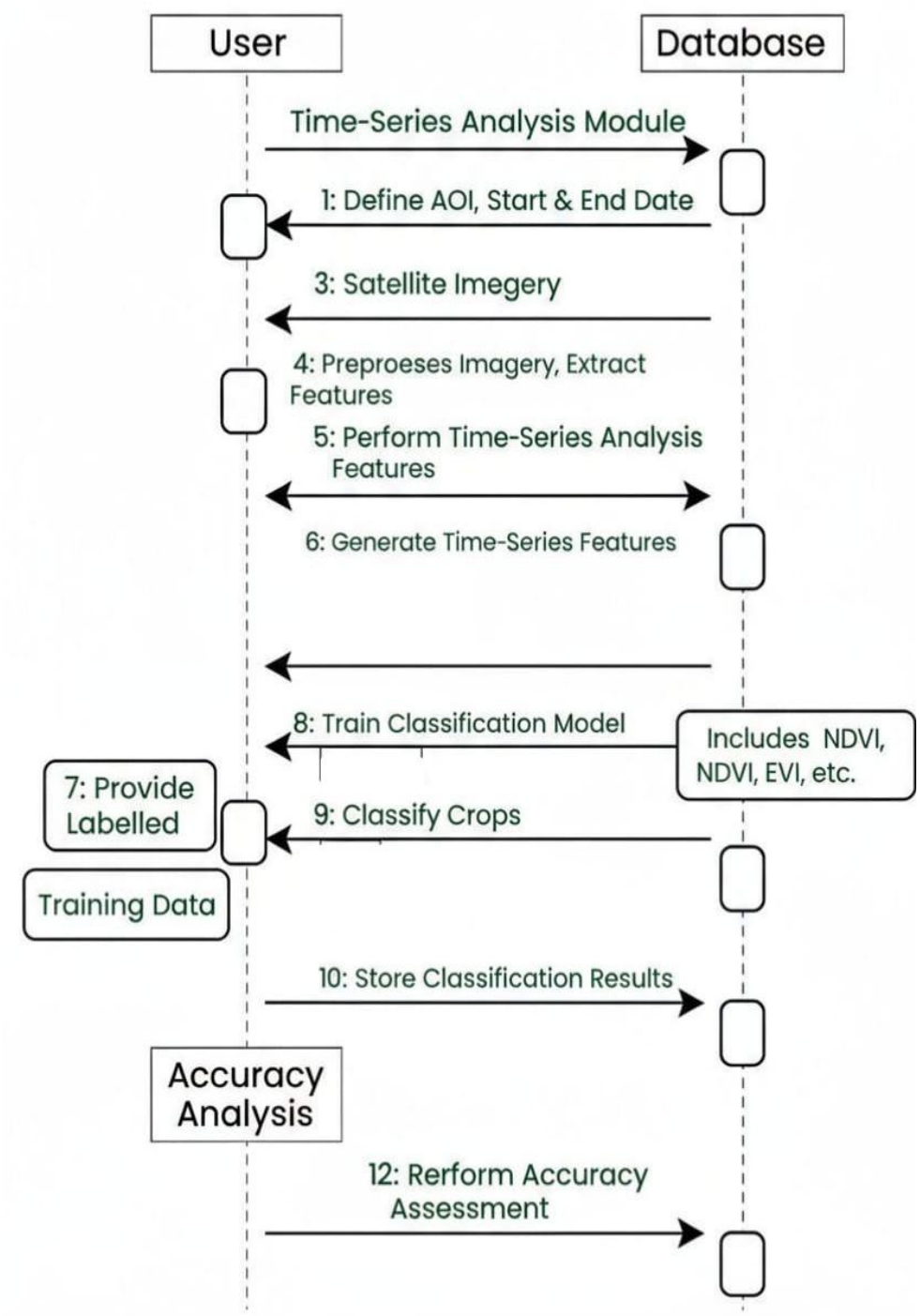


Level 2 DFD:

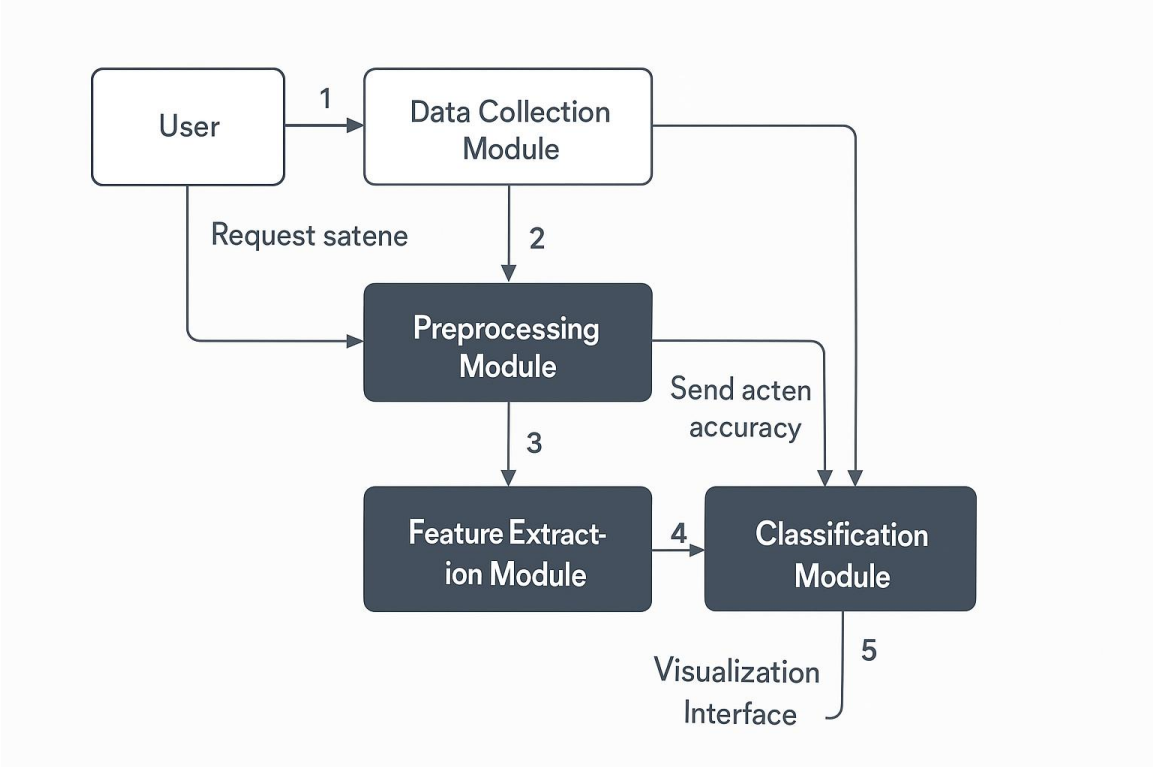

AOI Parameters and Dates
Input
AOI Boundary and Dates
AOI and Time Range
Raw
Images
2.1 Cloud Masking
Cloud-Free Images
Data Collection Module
Raw
Images
Pre-Processing Module
Cloud-Free Images
2.2 Atmospheric Correction
Pre-process Images
Pre-processed Imag-> Database
Surface Reflectance Images
Pre-processed Images Database
Pre-processed Extraction Module
Normalized Images
2.4 Vegetation Index Calculation
Normalized Images
Index Layers (NDVI, etc.)
AOI Boundary
2.5 Geographic Clipping
Clipped, Pre-processed Images
Feature Extraction Module

4.3 Behavioural Design

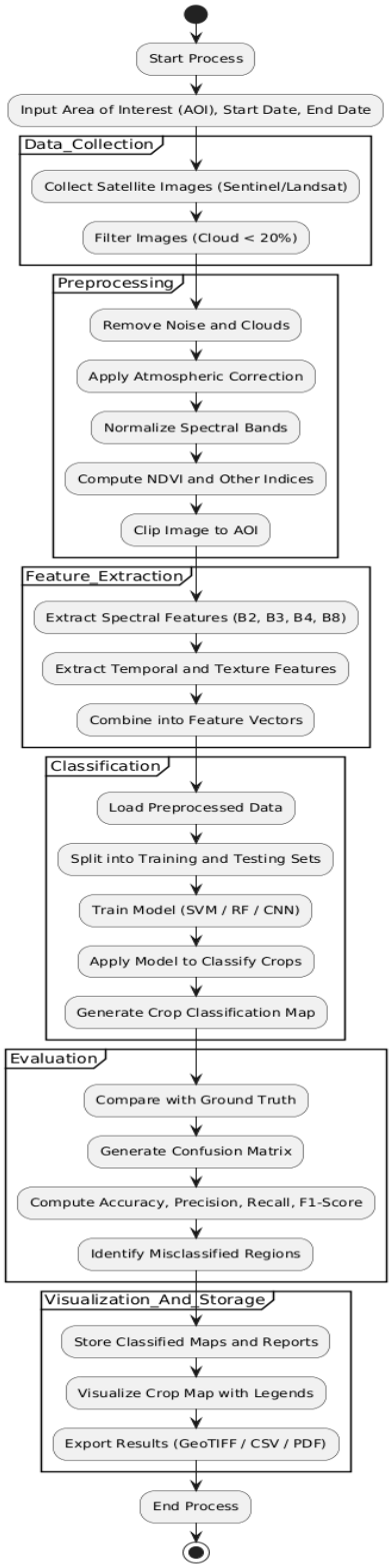
a. Sequence Diagram



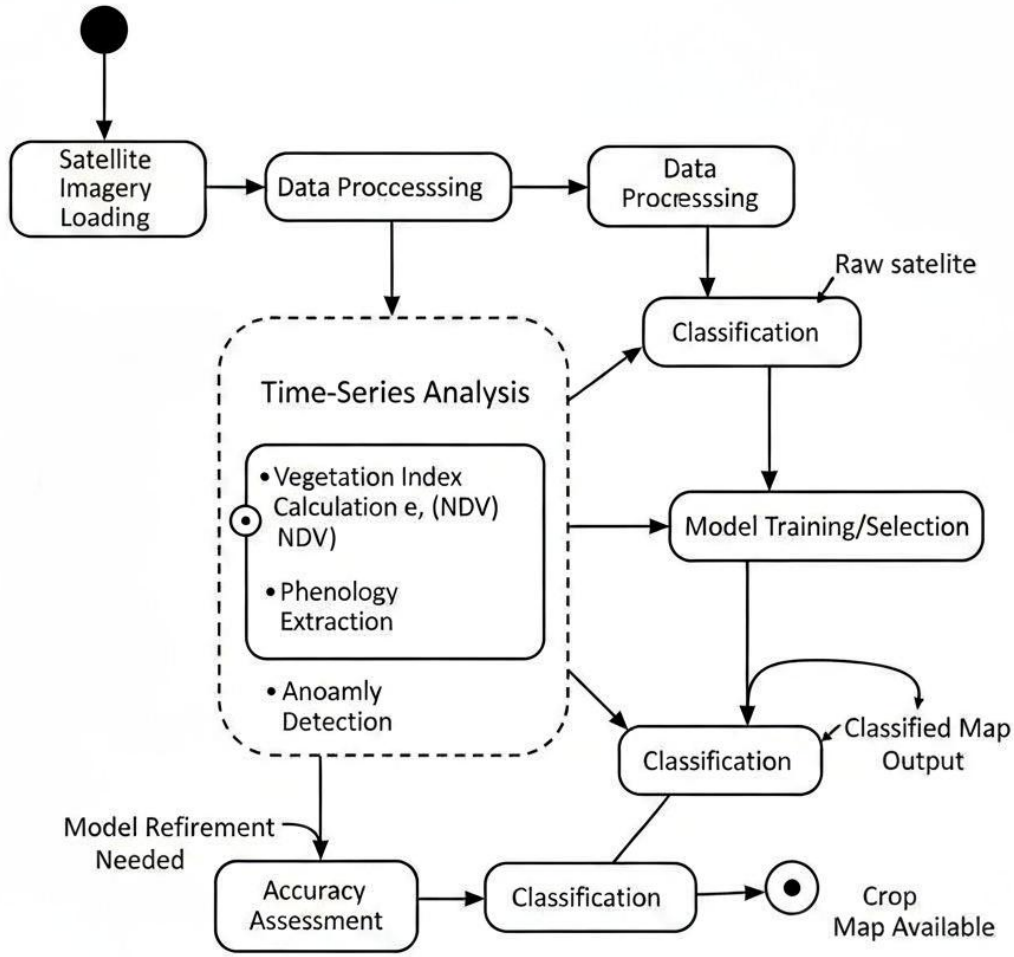
b. Collaboration diagram



c. Activity Diagram

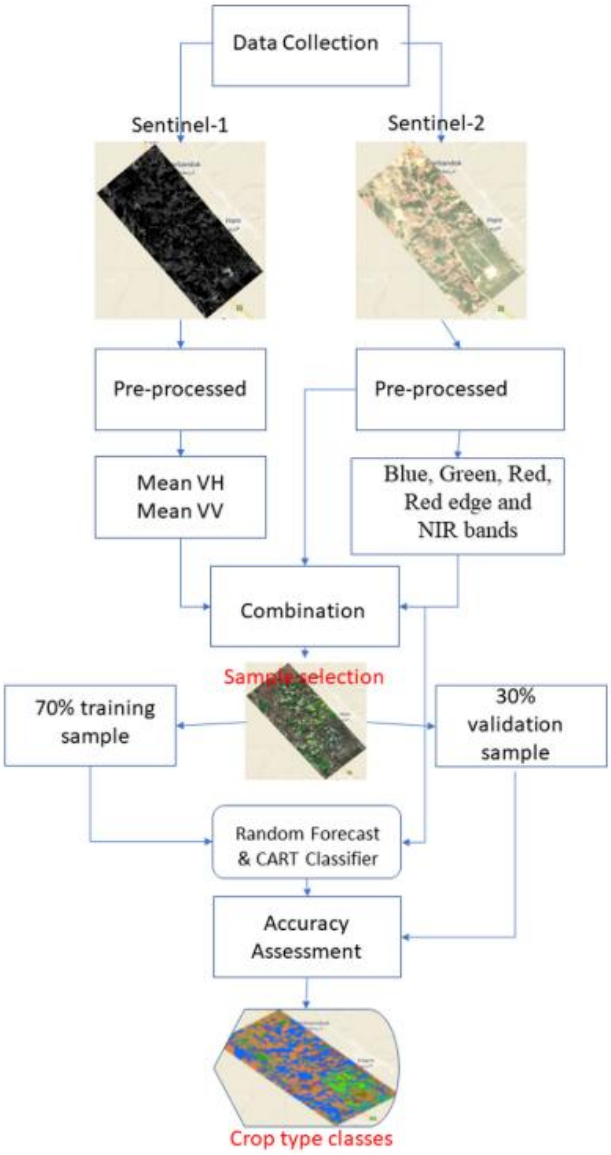


d. State-chart diagram



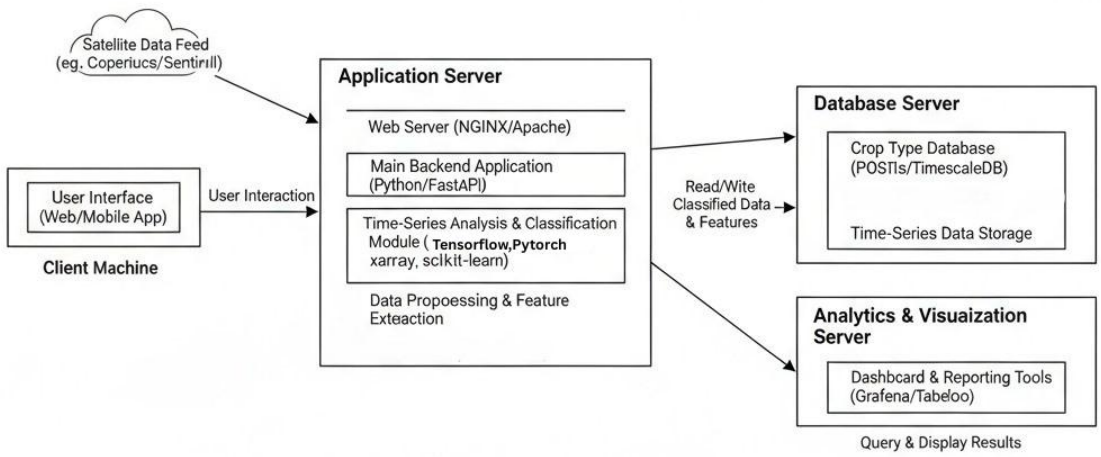
4.3. Implementation Design

a. Component Diagram



4.4. Environment Design

a. Deployment Diagram



5. Algorithms for Functional Requirements

1: Data Collection Module

Objective: Acquire multispectral satellite images for a given region and time range.
Input: Area of Interest (AOI), Start Date, End Date
Output: Preprocessed image collection (median composite)

Steps:

1. Define the Area of Interest (AOI) coordinates.
2. Set the time period for the desired crop season.
3. Access the Sentinel-2 or Landsat image collection from Google Earth Engine (GEE).
4. Filter images based on:
    - AOI
    - Date range
    - Cloud coverage < 20 %.
5. Generate a **median composite image** to minimize cloud and atmospheric noise.
6. Return the composite image for preprocessing.

2: Pre-processing Module

Objective: Prepare satellite imagery by removing noise and enhancing quality.
Input: Raw composite satellite image
Output: Clean, analysis-ready image

Steps:

1. Perform **cloud masking** using QA60 or cloud probability bands.
2. Apply **radiometric and atmospheric correction** (e.g., surface reflectance).
3. Normalize spectral bands to reduce brightness differences.
4. Calculate vegetation indices (e.g., NDVI = (NIR – RED)/(NIR + RED)).
5. Clip image to AOI boundaries.

6. Store preprocessed image for feature extraction.

3: Feature Extraction Module

Objective: Derive relevant spectral, temporal, and textural features for crop identification.
Input: Preprocessed image collection
Output: Feature vectors for each pixel or region

Steps:

1. Select key spectral bands: B2 (Blue), B3 (Green), B4 (Red), B8 (NIR).
2. Compute vegetation indices such as NDVI, EVI, and SAVI.
3. Extract temporal features (average, standard deviation over time).
4. Generate texture measures (entropy, contrast) from NDVI layer.
5. Combine all computed features into a feature vector dataset.
6. Store feature vectors in the feature database.

4: Classification Module

Objective: Classify crop types using machine learning or deep learning techniques.
Input: Feature vectors, labeled training data
Output: Crop classification map

Steps:

1. Load ground-truth samples or field survey data with crop labels.
2. Split dataset into **training (70 %)** and **testing (30 %)** sets.
3. Select and initialize a classifier (e.g., Random Forest, SVM, CNN).
4. Train the model using training features and crop labels.
5. Apply the trained model to unclassified satellite image pixels.
6. Generate a **classified image** assigning crop types to each pixel.
7. Store classification output for evaluation.

5: Evaluation Module

Objective: Assess the accuracy and performance of the classification results.
Input: Classified image, ground-truth data
Output: Accuracy metrics and confusion matrix

Steps:

1. Compare predicted crop labels with validation ground-truth samples.
2. Generate a **confusion matrix** with actual vs. predicted classes.
3. Compute accuracy metrics:
    - Overall Accuracy
    - Precision
    - Recall
    - F1-Score
    - Kappa Coefficient
4. Identify misclassified regions for model improvement.
5. Display performance metrics and classification map to the user.
6. Export final results to GeoTIFF or report format.

6: Data Storage and Visualization

Objective: Store processed data and visualize outputs for end-users.
Input: Classified maps, evaluation results
Output: Visualized crop map and stored records

Steps:

1. Save all images and results to the local or cloud database.
2. Visualize classified maps with distinct color palettes per crop.
3. Overlay crop boundaries, AOI, and NDVI on the map.
4. Provide download/export options (GeoTIFF, CSV, or PDF report).
5. End process.

6. References

• Dharmendra Singh, Parmod Kumar, 'Mapping of Rice Crop using Synthetic Aperture Radar (SAR) Data'.

• Sherrie Wang, Mutlu Ozdogan, 'Field-Scale Rice Area and Yield Mapping in Sri Lanka with Optical Remote Sensing and Limited Training Data', Remote Sens. 2025.

• Megharani Mayani, Rajeshwari Itagi, 'Sugarcane acreage estimation using satellite imagery and machine learning'.